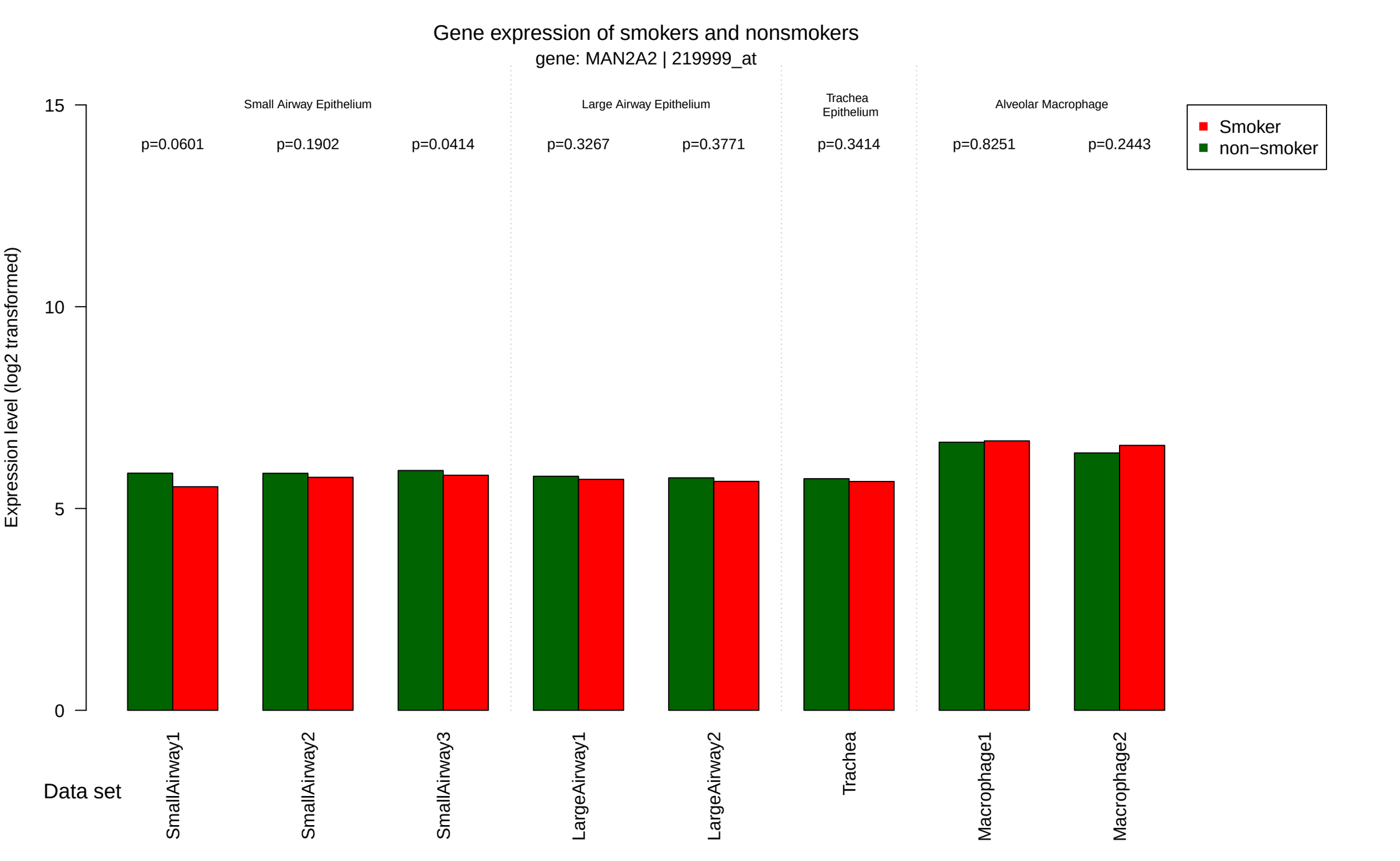
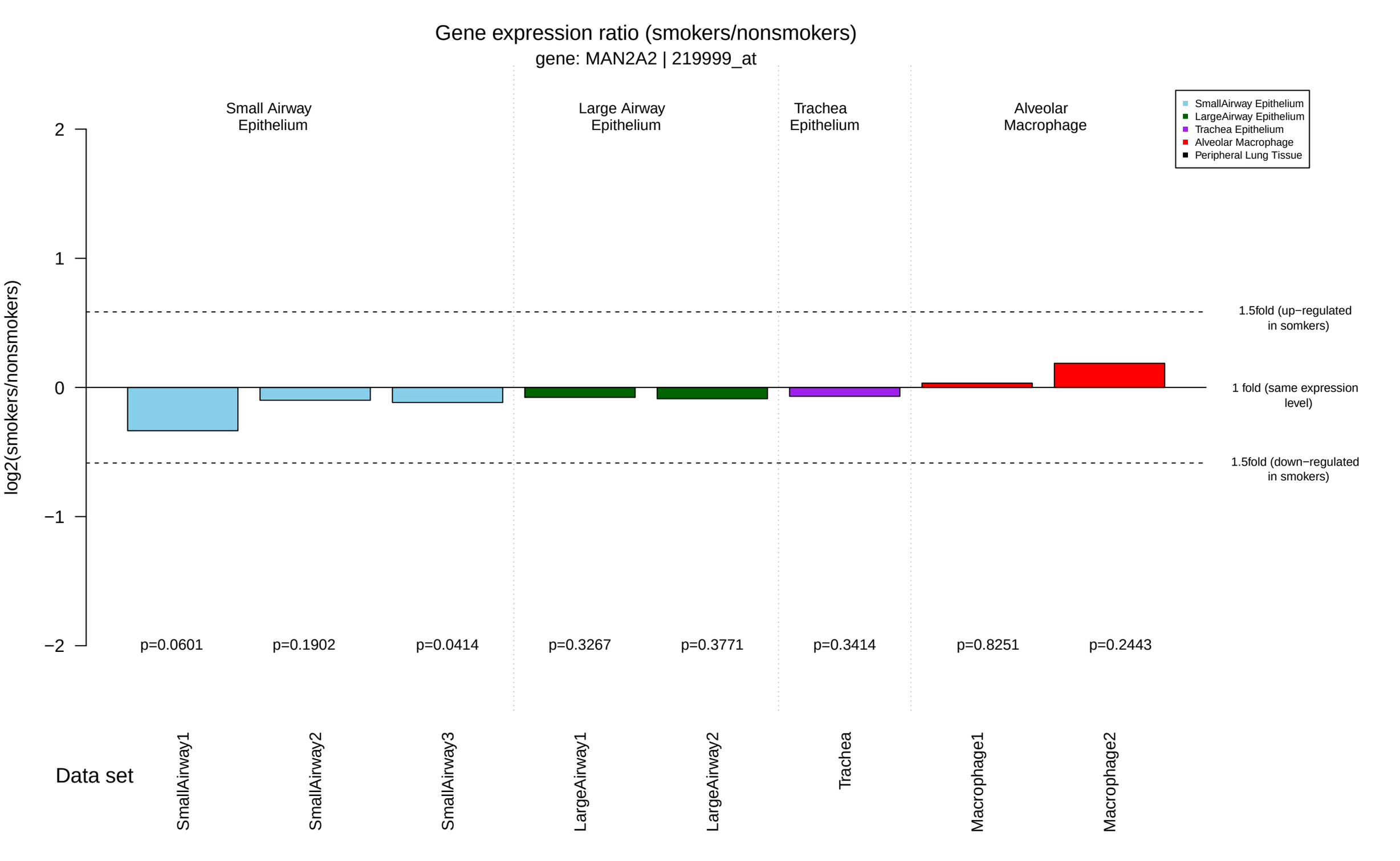
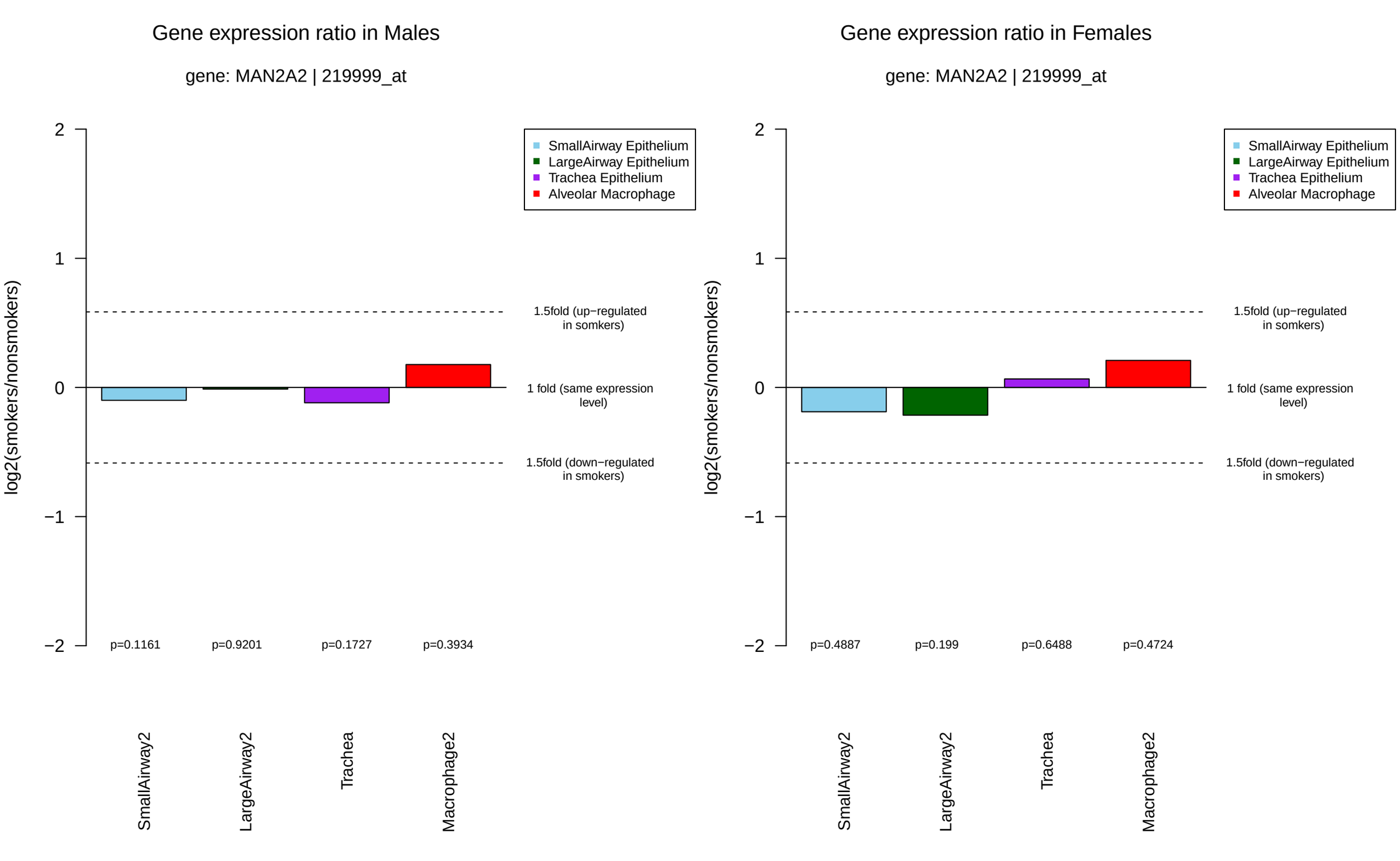
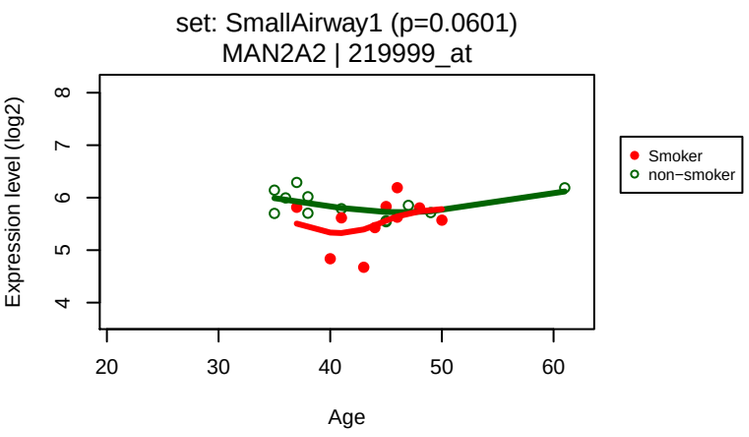
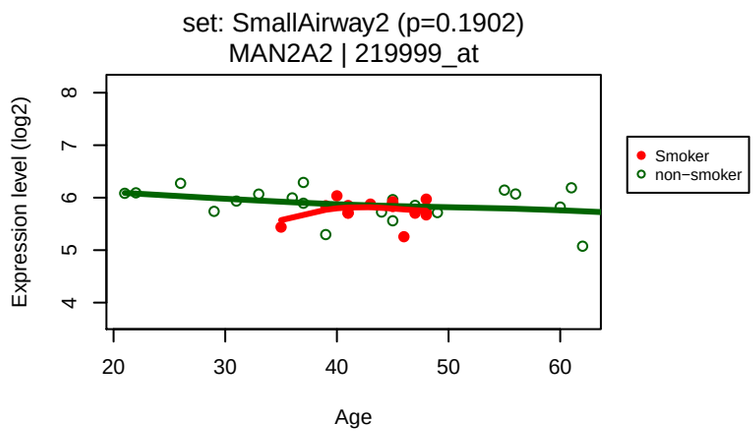
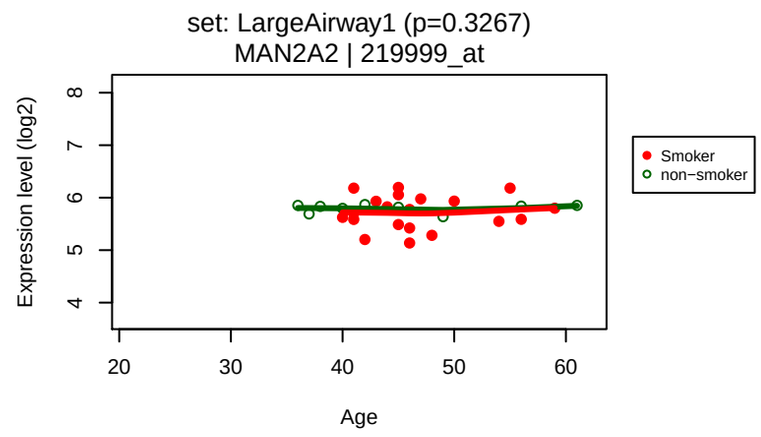

Gene expression of smokers and nonsmokers
gene: MAN2A2 | 219999_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0601
p=0.1902
p=0.0414
p=0.3267
p=0.3771
p=0.3414
p=0.8251
p=0.2443
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

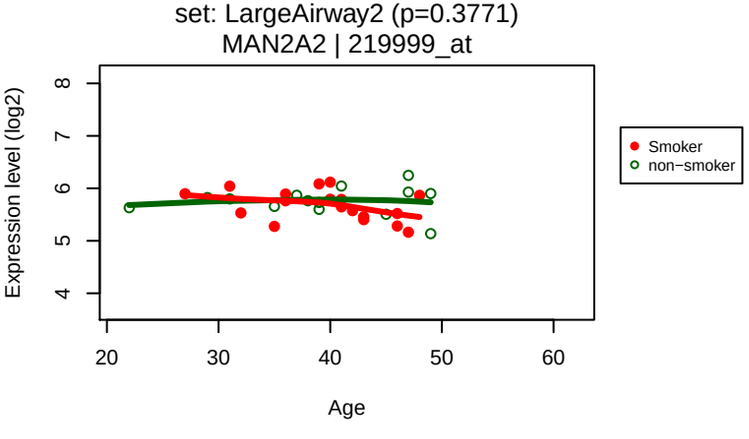

Gene expression ratio (smokers/nonsmokers)
gene: MAN2A2 | 219999_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0601
p=0.1902
p=0.0414
p=0.3267
p=0.3771
p=0.3414
p=0.8251
p=0.2443
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

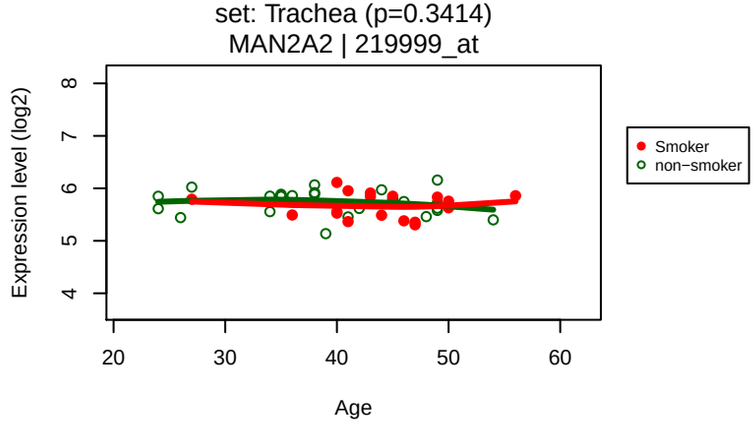

Gene expression ratio in Males
gene: MAN2A2 | 219999_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1161
p=0.9201
p=0.1727
p=0.3934
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MAN2A2 | 219999_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4887
p=0.199
p=0.6488
p=0.4724
SmallAirway2
LargeAirway2
Trachea
Macrophage2

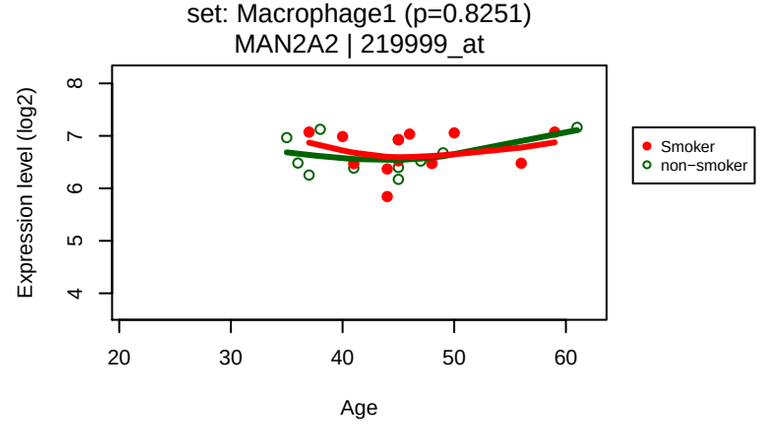

set: SmallAirway1 (p=0.0601)
MAN2A2 | 219999_at
Expression level (log2)
Age
Smoker
non−smoker

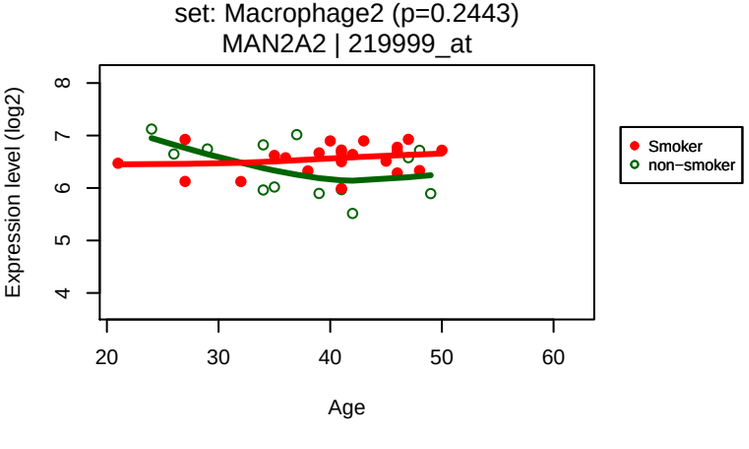

set: SmallAirway2 (p=0.1902)
MAN2A2 | 219999_at
Expression level (log2)
Age
Smoker
non−smoker

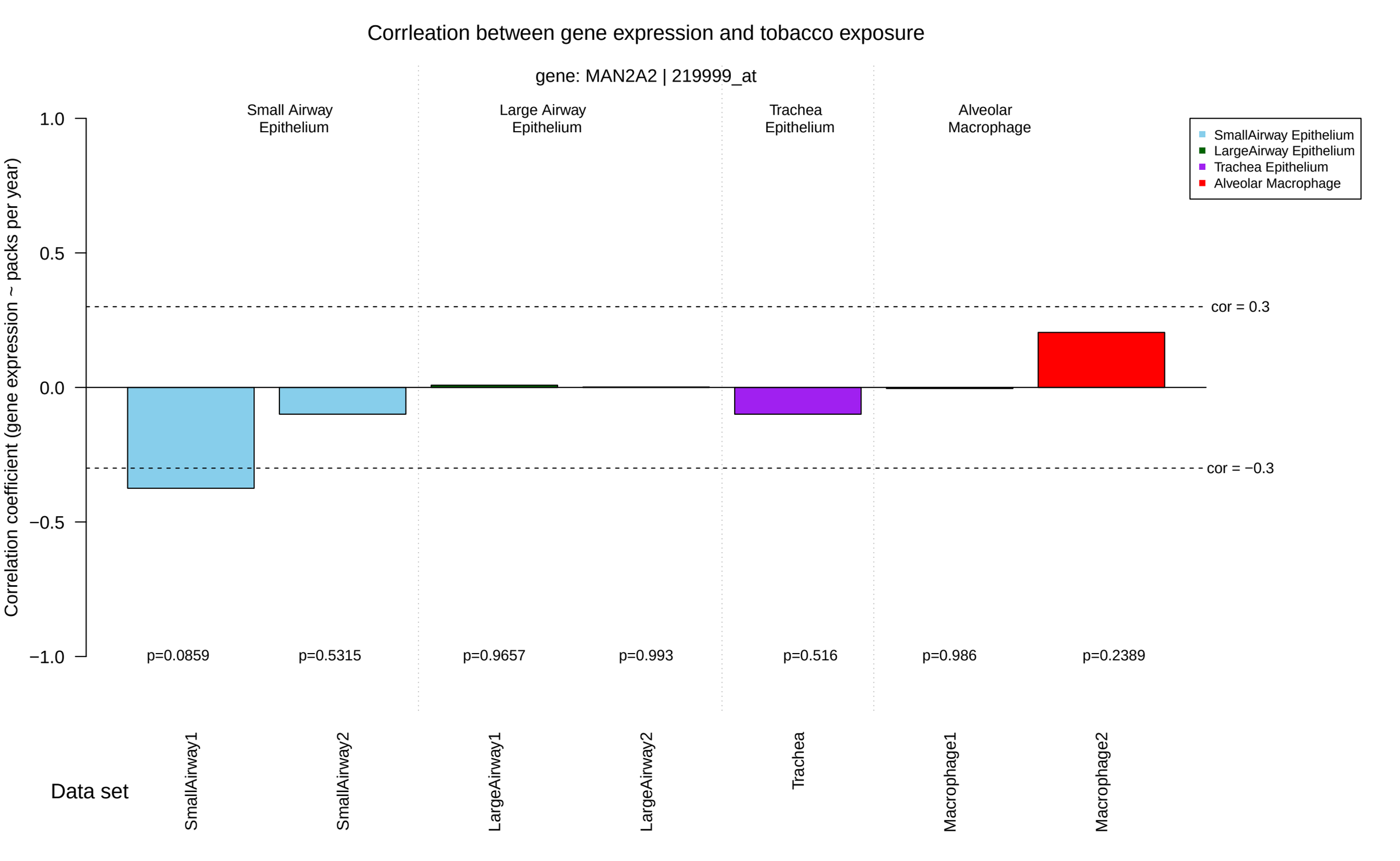

set: LargeAirway1 (p=0.3267)
MAN2A2 | 219999_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3771)
MAN2A2 | 219999_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3414)
MAN2A2 | 219999_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8251)
MAN2A2 | 219999_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2443)
MAN2A2 | 219999_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MAN2A2 | 219999_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0859
p=0.5315
p=0.9657
p=0.993
p=0.516
p=0.986
p=0.2389
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2